Augusto B. Corrêa

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SUMMARY

Research scientist working on **AI planning and LLM reasoning**. I develop methods that use search, program synthesis, and iterative learning to improve how LLMs reason and plan.

In recent work, I use LLMs to generate and refine executable programs, with automatic evaluation and counterexample-guided repair loops. These methods **outperform decades of hand-crafted algorithms** (NeurIPS 2025) while **reducing inference-time compute by up to 25**× (NeurIPS 2026).

I also showed that **iterative fine-tuning on curated agent trajectories implements reinforcement learning with an implicit reward function**, producing 196–401% improvements in planning capability, and **built the associated training pipeline over nearly 600,000 agent traces**.

My background in search and planning (IPC 2023 winner, ICAPS Outstanding Dissertation Award) gives me a foundation for studying inference-time search, learning from agent trajectories, and long-horizon reasoning in LLMs. I have 20 peer-reviewed papers at NeurIPS, AAAI, IJCAI, ICAPS, and Communications of the ACM (to appear). My work has been featured in the **Stanford AI Index Report 2026**.

EXPERIENCE

Junior Research Fellow 11/2024 – present

Christ Church, University of Oxford, UK

- Developed **inference-time search** methods in which LLMs generate executable candidate programs and an external evaluator tests and selects them. The selected heuristic functions **outperform decades of hand-crafted alternatives inside a real planner** (NeurIPS 2025) and extend to hierarchical planning (NeurIPS 2026).
- Replaced score-based sampling with **counterexample-guided repair**: the evaluator returns a concrete counterexample and the LLM revises the program. This **cuts LLM calls by up to 25**× and evaluation compute about 1000× while solving more tasks (NeurIPS 2026).
- Built the end-to-end post-training pipeline on close to **600,000 curated agent traces** (under review): trace generation and curation, LoRA/SFT, distributed training on GPUs, checkpoints, evaluation, and debugging failed jobs.
- Proved that **iterative fine-tuning on curated agent traces implements reinforcement learning with an implicit reward function**, with 196–401% gains in planning capability (arXiv:2512.24940); later generations find much longer plans than the base model. The mechanism is an alternative to explicit RL that relies on data curation rather than explicit rewards.
- Evaluated frontier-model reasoning against state-of-the-art planners: frontier LLMs are becoming competitive, and outperform the strongest planner baseline (first author, **Communications of the ACM**, to appear).

Doctoral Researcher 01/2020 – 10/2024

University of Basel, Switzerland · Supervisor: Prof. Malte Helmert

- Designed search algorithms, heuristic functions, and successor generation for **Powerlifted**, a C++/Python planner on first-order representations; maintained it as open-source software, reviewing contributions and managing releases.
- Built large-scale evaluation pipelines for search systems: benchmarks, repeated experiments, statistical comparisons, and failure analysis.
- **Won the International Planning Competition 2023** with Scorpion Maidu and Levitron, planner variants built on Powerlifted.
- Ph.D. awarded *summa cum laude*; thesis won the **ICAPS Outstanding Dissertation Award** (2025).

Research Assistant 02/2018 – 12/2019
University of Basel, Switzerland

- Developed the first lifted successor generation algorithm using query optimization (ICAPS 2020).
- Published on potential heuristics for classical planning (ICAPS 2019, IJCAI 2019).

EDUCATION

Ph.D. in Computer Science, University of Basel 2020 – 2024
Summa cum laude. **ICAPS Outstanding Dissertation Award** (2025).

M.Sc. in Computer Science, University of Basel 2018 – 2019
Major in Machine Intelligence.

B.Sc. in Computer Science, Federal University of Rio Grande do Sul 2013 – 2018
Best Bachelor's Thesis Award (Brazilian Computer Society).

SKILLS

LLM training	PyTorch, TensorFlow, Transformers, LoRA/SFT, PEFT, TRL, distributed GPU training
LLM reasoning	inference-time search, program synthesis, tool use, evaluation
Systems	Python, C++, Bash, Linux, Git, Slurm/HPC, Google Cloud
Languages	English, Portuguese, German, Spanish

SELECTED PUBLICATIONS

Full list: Google Scholar.

1. **Corrêa, A.B.**; Pereira, A.G.; and Seipp, J. *Classical Planning with LLM-Generated Heuristics: Challenging the State of the Art with Python Code.* **NeurIPS 2025.**
   Shows that LLM-generated heuristics outperform state-of-the-art classical planners.
2. Pereira, A.G.; **Corrêa, A.B.**; and Seipp, J. *Property-Guided LLM Program Synthesis for Planning.* **NeurIPS 2026**, to appear.
   Introduces counterexample-guided refinement of LLM-generated programs, improving solution quality while cutting LLM calls by up to 25× and evaluation compute by about 1000×.
3. Meneguzzi, F.; Buchweitz, A.; **Corrêa, A.B.**; Putrich, V.S.; and Pereira, A.G. *Hierarchical Task Network Planning with LLM-Generated Heuristics.* **NeurIPS 2026**, to appear.
   Extends LLM-generated program search to hierarchical, long-horizon problems, outperforming hand-engineered heuristic methods.
4. **Corrêa, A.B.**; Gelberg, Y.; Melo, L.C.; Shumailov, I.; Pereira, A.G.; and Gal, Y. *Iterative Deployment Improves Planning Skills in LLMs.* arXiv:2512.24940.
   Shows that iterative fine-tuning on selected agent trajectories implements reinforcement learning with an implicit reward, improving planning capability by 196–401%.
5. **Corrêa, A.B.**; Pereira, A.G.; and Seipp, J. *Frontier Large Language Models Are Becoming Competitive with State-of-the-Art Planners.* **Communications of the ACM**, accepted; to appear.
   First paper to show that end-to-end planning with LLMs is competitive with established symbolic planners.
6. **Corrêa, A.B.**; Dold, S.; and Helmert, M. *Potential Heuristics as Real-Valued Multilinear Polynomials.* **ICAPS 2026. Best Paper Runner-Up Award.**
   Recasts potential heuristics as multilinear polynomials, connecting classical planning to the analysis of Boolean functions.

SELECTED AWARDS AND SERVICE

- **Google's Gemini Academic Program Award** (2026): USD 10,000 for research on LLMs for planning.
- **ICAPS Outstanding Dissertation Award** (2025); **IPC Winner** (2023).
- **ICAPS Outstanding Program Committee Member Award** (2026).
- **ICAPS 2026 Best Paper Runner-Up**; **ICAPS 2023 Best Student Paper Runner-Up**; **ICAPS 2022 Best System Demonstration Award**.
- **Expert panel**: Transformative AI expert, Longitudinal Expert AI Panel (LEAP), Forecasting Research Institute (since 2026).
- **Workshop organizer**: LM4Plan (ICAPS 2025, ICAPS 2026, ICML 2026).
- **Programme committee**: NeurIPS, AAAI, IJCAI, ICAPS, KR, ECAI, SoCS.
- **Supervision**: 8 completed student theses (2 M.Sc., 6 B.Sc.).